

ECN 712 Fall 2025 EE Schlee
Problem Set IV Due Thursday, 19 Sept.

1. WARP for $L = 2$.¹

- (a) MWG, Exercise 2.F.14. (Assume that the demands satisfy budget balance.)
 - (b) Suppose that $L = 2$. Show that if a C^1 mapping $(p, w) \mapsto f(p, w)$ from positive price-wealth pairs into $\mathbb{R}_+^2$ satisfies budget balance and the weak axiom of revealed preference, then the 2×2 matrix S whose (i, j) element is $s_{ij} = \partial f_i / \partial p_j + (\partial f_i / \partial w) f_j$ is a *symmetric* matrix for every $(p, w) \gg 0$. (Hint: use part (a) and Corollary 4—substituting f for d —from my notes on Demand Properties. You *should* understand how to derive Corollary 4 for f using budget balance and homogeneity, but you can simply *use* Corollary 4 applied to f for this part for now.)
2. MWG, 3.G.4, (b) & (c). You should allow for corner solutions in part (c), but *do* assume that the u_ℓ functions are differentiable with $u'_\ell > 0$. [If you get stuck, start with $L = 2$ and assume interiority, then generalize.] To clarify: for single-valued demands, goods are *normal* if, for every $w_1 > w_0 \geq 0$ and $p \gg 0$, $d(p, w_1) > d(p, w_0)$. The conclusion for the compensated demands follows immediately from the conclusion for the ordinary demands. (d) Can you think of a *non-pathological* example of 3 goods whose conditional preferences are not representable by an additively separable function?
3. Consider an economy with I consumers and let consumer i have an indirect utility function of the Gorman form $v^i(p, w^i) = a^i(p) + b(p)w^i$, where w^i is i 's wealth, for $i = 1, \dots, I$. (The functions take this form on an open neighborhood of some price-wealth pair $(\bar{p}, \bar{w}^i) \gg 0$.) Assume that $a^i(\cdot)$ and $b(\cdot)$ are differentiable functions and that $b(\cdot)$ is a positive function (so Roy's Identity holds).
- (a) What restrictions on consumer i 's indirect utility function *are consistent with* preferences that are (i) homothetic; (ii) quasilinear with respect to good L ?
 - (b) Show that aggregate demand in this economy depends simply on aggregate wealth and not its distribution. Verify that the sum $\sum v^i(p, w^i)$ just depends on aggregate wealth and that the “aggregate indirect utility function” $V(p, \sum w^i) = \sum v^i(p, w^i)$ indeed generates the economy's aggregate demand. Does the “aggregate” expenditure function $E(p, \sum u^i) = \sum e^i(p, u^i)$ satisfy the properties that individual expenditure functions must (Theorem 1 in my notes on testable implications of consumer theory)?
 - (c) Show that if the “representative consumer”—a hypothetical consumer with wealth $\sum w^i$ and indirect utility V from part (b)—weakly prefers price vector $q \gg 0$ to price vector $p \gg 0$, then there is a way to redistribute wealth among consumers so that they all weakly prefer q with the new wealth to p with the old. (You may ignore nonnegativity restrictions on wealth in your redistribution.)
 - (d) **Optional.** Problem 3 from the Fall 2022 midterm (available on Canvas; my notes on the problem are also on Canvas). It illustrates that aggregate demand can be Cobb-Douglas, but the family of indirect utilities is not Gorman, and the conclusion from part (c) fails.
4. Suppose Y exhibits *nondecreasing returns to scale*. *Evaluate*: “The profit function takes at most two values, 0 or $+\infty$ ” (or, if you prefer, it either equals 0 or problem (PM) has no solution).

¹Putting problems 1a and 1b together: for $L = 2$, budget balance and the weak axiom imply that the demands are homogeneous of degree 0 in (p, w) ; and that the Slutsky substitution matrix is symmetric and negative semidefinite (the last fact we proved in class). By the argument in MWG, Section 3.H, pp. 78-80, there is therefore a utility function that generates these demands as the solution to problem (UM).